MCQ solutions

1. **B.** The fundamental theorem gives $F(b) - F(a)$. It applies to an antiderivative of f, not to f itself.

2. **C.** For $a < b$, the integral keeps the sign of a function that is negative or zero. The geometric area, for its part, is positive or zero.
3. **A.** The integral is divided by the length of the interval $b - a > 0$. For a constant function, we recovers this constant.
4. **B.** We calculate the areas on the intervals where the sign is constant and add their positive values. Without splitting, areas can cancel each other out.
5. **B.** The derivative of e^x is e^x. The other proposed functions have different derivatives.
6. **A.** The usual integration by parts formula applies to two functions of class $\mathcal{C}^1$ on the interval. This framework guarantees the continuous derivatives and the integrals used.
7. **A.** The formula comes from $(uv)' = u'v + uv'$. Integrating and then isolating the first term gives $\int_a^b u'v = [uv]_a^b - \int_a^b uv'$.
8. **B.** An increasing function lies above the height taken at the left end of each interval. The left rectangles therefore give a lower bound.
9. **A.** For $x \in [0;1]$, $0 \leq x^n \leq 1$; one can multiply by $f(x) \geq 0$, then integrate. The bound may be refined by an upper bound of f.
10. **B.** The product of a flow rate in litres per minute and a duration in minutes is expressed in litres. The integral represents a cumulative volume.

True/false. False: $\int_0^1 (-1)\,\mathrm{d}x = -1$. True: the difference of two antiderivatives has zero derivative on an interval, so is constant. False: whether it increases or decreases determines which side gives the lower bound.

Solutions to the applications and guidance

Application 1. $g = 2f$: integral 3, area 5. The factor 2 is positive, so the signs are preserved and both areas are doubled.

Application 2. $A(x) = 1 - e^{-x}$, $A' = e^{-x} > 0$, hence increasing with limit 1. The area under e^{-t} between 0 and x increases and approaches 1.

Application 3. $m = 1/3$. The rectangle has width 1 and height 1/3. The mean of the endpoint values would be 1/2: it measures the area of the upper trapezium, not the area under the curve.

Application 4. $L_4 = \frac{1}{4}(0 + 1/16 + 1/4 + 9/16) = 7/32$; $U_4 = 15/32$. Monotonicity gives $L_4 \leq 1/3 \leq U_4$. $U_n - L_n = \frac{1}{n}[1^2 - 0^2] = 1/n$.

Application 5. Take u=x and $v = \ln x$, therefore $u' = 1$, $v' = 1/x$. The integral equals $[x \ln x]_1^e - \int_1^e 1\,\mathrm{d}x = e - (e-1) = 1$. An antiderivative is $x \ln x - x$, whose derivative is ln x.

Application 6. On [0;1], $0 \leq x^{n+1} \leq x^n$; multiply by $e^{-x} > 0$, then integrating proves positivity and decrease. From $e^{-1} \leq e^{-x} \leq 1$, we get the bounds, then $I_n \to 0$. Integration by parts applied to I_{n+1} gives $I_{n+1} = -e^{-1} + (n+1)I_n$. This relation is exact but is not enough on its own to calculate the limit.

Application 7. $f(x) = x$ works. If f is continuous and positive or zero, a strictly positive would, by continuity, give a strictly positive area on a small interval, contradiction. Thus a continuous nonnegative function with zero integral is identically zero.

Session plan

Keep setting up the signed area separate from integration by parts. Work on rectangles is done first on the drawing, then with sums. In the sequence of integrals, let students choose between monotone convergence and bracketing; the second gives the value of the limit directly here.

Teaching notes and solutions

Guidance for the teacher

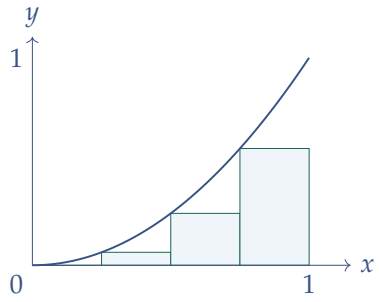
Start with a positive area, then show an example whose positive and negative areas cancel out. For a mean value, require a unit and an interpretation. The sequences of integrals come in S66; set aside a genuine research phase for choosing the comparison. The proof of the fundamental theorem is limited to the increasing positive case stated in the curriculum. Trigonometric calculations will be revisited after May.

Solutions to the activity

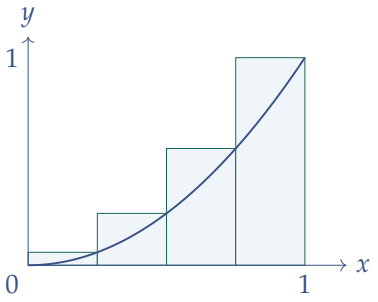
1. The four intervals have width $\frac{1}{4}$. The sum on the left is

$$L_4 = \frac{1}{4}(0^2 + (\frac{1}{4})^2 + (\frac{1}{2})^2 + (\frac{3}{4})^2) = \frac{7}{32}.$$

On the right, $U_4 = \frac{1}{4}((\frac{1}{4})^2 + (\frac{1}{2})^2 + (\frac{3}{4})^2 + 1) = \frac{15}{32}$. The heights of the rectangles are the values at the corresponding endpoints.



Left-hand rectangles: $L_4 = \frac{7}{32}$.



Right-hand rectangles: $U_4 = \frac{15}{32}$.

2. For $x \in [\frac{k}{4}, \frac{k+1}{4}]$, $(\frac{k}{4})^2 \leq x^2 \leq (\frac{k+1}{4})^2$ because the square function is increasing on $[0,1]$. Integration on the four intervals and then adding give $\frac{7}{32} \leq I \leq \frac{15}{32}$.

3.

```
def rectangles_carre(n):
    h = 1/n
    bas = h* sum((k*h)**2 for k in range (n))
    haut = h* sum((k*h)**2 for k in range (1, n+1))
    return bas, haut
```

The intermediate terms cancel out: $U_n - L_n = (\frac{1}{n})(1^2 - 0^2) = \frac{1}{n}$.

4. $\frac{1}{n} \leq 0{,}01 \iff n \geq 100$. One hundred subdivisions suffice; it is the first integer giving this width with these two sums.

5. An antiderivative of x^2 is $\frac{x^3}{3}$. Hence $I = [\frac{x^3}{3}]_0^1 = \frac{1}{3}$, indeed between $\frac{7}{32}$ and $\frac{15}{32}$.

E1. An antiderivative is $F(x) = x^3 - x^2 + x$. The integral equals $F(2) - F(0) = 8 - 4 + 2 = 6$.

E2. On $[0,1]$, $x - x^2 = x(1-x) \geq 0$, so the line lies above the parabola. The area is $\int_0^1 (x - x^2)\,\mathrm{d}x = [\frac{x^2}{2} - \frac{x^3}{3}]_0^1 = \frac{1}{6}$ unit of area.

E3. The length of the interval is 2. The mean value is $(\frac{1}{2})\int_0^2 e^x\,\mathrm{d}x = (\frac{1}{2})[e^x]_0^2 = \frac{e^2-1}{2}$.

E4. Let us choose $u = \ln x$, $u' = \frac{1}{x}$, $v' = 1$ and $v = x$ on $[1, e]$. Integration by parts gives $\int_1^e \ln x \, dx = [x \ln x]_1^e - \int_1^e 1 \, dx = e - (e-1) = 1$.

E5. On $[0,1]$, $1 + x \geq 1$ and $x^n \geq 0$, hence $0 \leq \frac{x^n}{1+x} \leq x^n$. Integration gives $0 \leq I_n \leq [\frac{x^{n+1}}{n+1}]_0^1 = \frac{1}{n+1}$. The squeeze theorem gives $I_n \to 0$.

Solutions to homework DM8

1. $I_0 = \int_0^1 e^x \, dx = e - 1$. For I_1, integration by parts with $u = x$, $v' = e^x$ gives $I_1 = [xe^x]_0^1 - \int_0^1 e^x \, dx = e - (e-1) = 1$.

2. For $n \geq 0$, let us choose $u = x^{n+1}$, $u' = (n+1)x^n$, $v' = e^x$ and $v = e^x$. The functions are differentiable with continuous derivatives on $[0,1]$. Then

$$I_{n+1} = [x^{n+1}e^x]_0^1 - (n+1)\int_0^1 x^n e^x \, dx = e - (n+1)I_n.$$

3. On $[0,1]$, $0 \leq x^{n+1} \leq x^n$. By multiplying by $e^x > 0$ then integrating, $0 \leq I_{n+1} \leq I_n$: the sequence is positive and decreasing (in the wide sense). It is even strictly positive and strictly decreasing, since the integrands of I_n and $I_n - I_{n+1}$ are strictly positive on $]0,1[$.

4. Since $1 \leq e^x \leq e$ and $x^n \geq 0$ on $[0,1]$, we can integrate $x^n \leq x^n e^x \leq e x^n$. Hence

$$\frac{1}{n+1} \leq I_n \leq \frac{e}{n+1}.$$

Both bounds tend to 0; the squeeze theorem gives $I_n \to 0$.

5. If $\widetilde{I}_n = I_n + \varepsilon_n$ and that the relationship is then applied without further error, then

$$\widetilde{I}_{n+1} - I_{n+1} = -(n+1)(\widetilde{I}_n - I_n) = -(n+1)\varepsilon_n.$$

The absolute value of the error is multiplied by $n+1$. In machine arithmetic, new errors add up at each step. The decrease of the exact integrals therefore does not guarantee stability of this method of calculation.

Solutions to the textbook or past-paper work

Exercise 28. On $[0;1]$, let us choose $u(x) = x$ and $v(x) = e^x$. These functions are of class $\mathcal{C}^1$, $u'(x) = 1$ and $v'(x) = e^x$. Integration by parts gives

$$\int_0^1 xe^x \, dx = [xe^x]_0^1 - \int_0^1 e^x \, dx = e - (e-1) = 1.$$

Exercise 29. On $[1;e]$, let us choose $u(x) = \ln x$ and $v(x) = \frac{x^2}{2}$. Then $u'(x) = \frac{1}{x}$ and $v'(x) = x$; these functions are of class $\mathcal{C}^1$. Hence

$$\begin{aligned}\int_1^e x \ln x \, dx &= \left[\frac{x^2}{2}\ln x\right]_1^e - \frac{1}{2}\int_1^e x \, dx \\ &= \frac{e^2}{2} - \frac{e^2-1}{4} = \frac{e^2+1}{4}.\end{aligned}$$